

Web Technologies Project Report: Tary Ethno Cafe (Astana Flagship)

Course: Web Technologies / Frontend Fundamentals

Repository URL: <https://github.com/jasulansam/web>

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1. Project Overview

- **Objective:** Development of a multi-page website for the Tary Ethno Cafe branch (Kabanbay Batyr 15/1, Astana) adhering strictly to semantic HTML5 standards without external styling or client-side scripting frameworks.
- **Core Architectural Rules:**
 - Strict landmark hierarchy (<header>, <nav>, <main>, <footer>).
 - Pure semantic structure (no inline styling, zero presentation attributes like border="1" or align).
 - Tabular data formatting using accessible table structures (<table>, <caption>, <thead>, <tbody>, and th scope).
 - Form inputs configured with explicit label associations, logical field grouping (<fieldset>, <legend>), and native HTML5 validation constraints.

2. Task A — Anatomy of a Real Page (Reference Audit: coffeeboom.kz)

2.1 Head Elements and Metadata Inspection

- **DOCTYPE:** <!DOCTYPE html>
- **Language:** lang="kk" in <html class="html" lang="kk">
- **Charset:** Legacy HTTP-EQUIV syntax <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
- **Viewport:** <meta name="viewport" content="width=device-width, initial-scale=1.0">
- **Title:** Coffee Boom – Қазақстан, Ресей және Өзбекстандағы кофеханалар желісі.
- **Three Key Meta Tags:**
 1. <meta name="Description" content="Coffee Boom – 2010 жылдан бері жұмыс істеп келе жатқан, 130-дан астам кофеханасы бар халықаралық желі...">
 2. <meta name="theme-color" content="whitesmoke">
 3. <meta property="og:image" content="https://coffeeboom.kz/wa-data/public/wcms/images/07/04/00/461/461.0x0.png">

2.2 DevTools Source Inspection (Div-Soup Demonstration)

Inspect Element - coffeeboom.kz (Div-Soup & Legacy Metadata)

```
<!DOCTYPE html>
<html class="html" lang="kk">
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Coffee Boom – Қазақстан, Ресей және Өзбекстандағы кофеханалар желісі.</title>
    <meta name="Description" content="Coffee Boom – 2010 жылдан бері жұмыс істеп келе жатқан...">
  </head>
  <body>
    <div class="site-wrapper">
      <div class="header-box"> <!-- Div used instead of <header> -->
        <div class="nav-menu-bar"> <!-- Div used instead of <nav> -->
          <div class="menu-item"><a href="/about">Біз туралы</a></div>
          <div class="menu-item"><a href="/franchise">Франшиза</a></div>
          <div class="menu-item"><a href="/roastery">Кофе фабрикасы</a></div>
        </div>
      </div>
      <div class="main-content-area"> <!-- Div used instead of <main> -->
        <div class="slider-container">...</div>
        <div class="cards-grid"> <!-- 570 generic <div> tags total -->
```

Element Type	Tag Name	Occurrence Count	Status / Evaluation
Semantic Header	<header>	1	Present for brand banner
Semantic Navigation	<nav>	0	Severe issue: navigation wrapped only in generic divs
Sectional Grouping	<section>	0	No thematic section tags used
Independent Article	<article>	0	Blog/magazine articles use plain divs
Complementary Aside	<aside>	0	No peripheral landmarks
Data Tables	<table>	0	No structured tables for menu or branch data
Generic Containers	<div>	570	Excessive div-soup throughout entire layout

3. Three Structural Mistakes and Our Solutions

1. Extreme "Div-Soup" and Missing Navigation Landmarks:

- *Mistake:* On coffeeboom.kz, the main navigation bar containing over 20 links is wrapped in generic `<div>` tags rather than a semantic `<nav>`. Screen readers and search engine crawlers cannot identify navigation landmarks.
- *Our Solution:* On all our pages (`index.html`, `menu.html`, `booking.html`, `colophon.html`), site navigation is strictly encapsulated in a semantic `<nav>` tag holding a structured `<ul>` list.

2. Outdated Legacy Charset Declaration:

- *Mistake:* The site uses the obsolete HTML4 syntax `<meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>`.
- *Our Solution:* We implemented the modern, concise HTML5 standard `<meta charset="UTF-8">` on every document.

3. Absence of Accessible Tables for Tabular Information:

- *Mistake:* The real website displays regional branches, locations, and menu pricing via unsemantic floating divs (zero `<table>` elements), making data extraction inaccessible for assistive tools.
- *Our Solution:* In our files (specifically `menu.html` for pricing), we implemented an accessible `<table>` complete with `<caption>`, `<thead>`, `<tbody>`, and explicit scopes (`scope="col"`, `scope="row"`).

4. Hand-Drawn Page Structure Sketch

As required by Assignment 1, the page structure of the audited competitor website was analyzed and sketched on paper, noting semantic mistakes, structural landmarks, and signed by the student.



5. Task B — Reflective Essay

Reflective Essay: Jasulan

A web page is fundamentally a plain-text document structured with HyperText Markup Language (HTML). At its core, an HTML file consists of human-readable elements demarcated by tags, attributes, text nodes, and entities. When a user navigates to a local HTML file or web address, the browser receives a continuous stream of raw bytes.

The rendering engine decodes these bytes into characters using the declared character encoding (UTF-8), tokenizes the stream into distinct start tags, end tags, and content, and constructs the Document Object Model (DOM) tree. Concurrently, the engine parses external stylesheets to generate the CSSOM (CSS Object Model). The browser merges DOM and CSSOM into a Render Tree, filtering out non-rendered nodes (such as `<head>` or elements with `display: none`). Next, the layout engine executes geometry calculations to determine the precise size and coordinates of each element on the viewport. Finally, the painting stage rasterizes vectors, text, borders, and colors onto physical pixels displayed on the screen. Because our assignment strictly excludes CSS and JavaScript, the browser renders directly using its default User-Agent stylesheet.

In my codebase, I strictly prioritized semantic markup over generic `<div>` containers:

1. In `menu.html`, I wrapped our pricing data inside an accessible `<table>` with `<thead>` and `<tbody>` instead of styled grid divs, giving assistive screen readers clear tabular data relationships.
2. In `booking.html`, I utilized `<fieldset>` paired with `<legend>` rather than wrapper divs to logically group reservation preferences, ensuring screen readers announce category contexts.
3. In `index.html`, I chose `<article>` instead of `<div>` for the cafe heritage section because it forms an independent, self-contained guide that can stand on its own.

When a visitor clicks the submit button on `booking.html` today, the browser triggers native client-side constraint validation against attributes like `required`, `type="email"`, `type="number"`, and `type="tel"`. Because no server-side backend script processes the request, the browser attempts an HTTP POST request to `action="#"`, serializing form values and reloading the static page with fragment anchor.

6. AI Assistance & Usage Logs

Interactive logs documenting prompt history, academic integrity declarations, and AI-assisted workflow iterations are maintained in the GitHub repository:

Jasulan — AI Usage Log: <https://github.com/jasulansam/web/blob/main/ai-log.md>

7. W3C Validation Evidence

All four project documents were submitted to the official W3C Nu HTML Validator, passing with zero errors and zero warnings.

1. val-index (index.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Official website of Tary Ethno Cafe in Astana, Kazakhstan, celebrating authentic Kazakh gastronomic tr
  <meta name="author" content="Jasulan">
  <!-- Author: Jasulan -->
  <title>Tary Ethno Cafe Astana - Authentic Kazakh Coffee & Heritage</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

[About this checker](#) • [Report an issue](#)

2. val-menu (menu.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Discover the artisanal menu at Tary Ethno Cafe Astana, offering signature millet beverages, steppe tea
  <meta name="author" content="Jasulan">
  <!-- Author: Jasulan -->
  <title>Menu & Traditional Offerings - Tary Ethno Cafe Astana</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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3. val-booking (booking.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Reserve a table at Tary Ethno Cafe in Astana for authentic tea ceremonies, specialty coffee tastings, and live music performances.">
  <meta name="author" content="Jasulan">
  <!-- Author: Jasulan -->
  <title>Table Reservations & Inquiries - Tary Ethno Cafe Astana</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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4. val-colophon (colophon.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Colophon and technical documentation explaining how the Tary Ethno Cafe website was constructed using modern web technologies.">
  <meta name="author" content="Jasulan">
  <!-- Author: Jasulan -->
  <title>Colophon & Technical Architecture - Tary Ethno Cafe</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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8. HTML5 Semantic Elements Checklist

The comprehensive line-by-line checklist of every required HTML5 semantic tag, attribute, list, and table structure is documented in the repository:

URL: <https://github.com/jasulansam/web/blob/main/tag-checklist.md>